

Aryana Haghjoo

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Aryana Haghjoo is an early-career astrophysicist with a strong foundation in both theoretical and data-driven research. She holds an M.Sc. in Physics from McGill University, where she specialized in computational global 21cm cosmology, and is currently pursuing a Ph.D. in Astronomy at the University of California, Riverside. Her recent research leverages deep learning to super-resolve low-resolution galaxy spectra.

Education

Ph.D. Astronomy, [University of California, Riverside \(UCR\)](#)

GPA: 3.7/4

Sept 2025 – Jun 2030 (Expected)

Riverside, California, USA

M.Sc Astronomy, [UCR](#)

GPA: 3.7/4

Sept 2024 – Sept 2025

Riverside, California, USA

M.Sc Physics, [McGill University](#)

GPA: 3.5/4

Sept 2021 – Dec 2023

Montreal, Quebec, Canada

Thesis Title: [Parameter Estimation of The Global 21cm Signal](#)

B.Sc Physics, [Sharif University of Technology \(SUT\)](#)

GPA: 18.25/20

Sept 2018 – Jul 2021

Tehran, Iran

Publications

14 publications, 75 citations (NASA ADS), *h*-index: 5.

First-Author Manuscripts

- **Haghjoo, A.**, Hemmati, S., Mobasher, B., Chartab, N., de la Vega, A., Eifler, T., Ramey, E., Nayyeri, H., & Sattari, Z. *Learning to See Sharper: A Physics-Informed Artificial Intelligence Framework for Super-Resolving Galaxy Spectra*. Under review at the *Astrophysical Journal*.
- **Haghjoo, A.**, Hemmati, S., & Mobasher, B. *Sharper at What Cost? Benchmarking Deep Learning Against Classical Deconvolution for Galaxy Spectral Super-Resolution*. In preparation for submission to the *Astrophysical Journal*.

Selected Co-Author Publications

- Abedini, F., Gozaliasl, G., Zonoozi, A. H., **Haghjoo, A.**, et al. *COSMOS-Web: Estimating Physical Parameters of Galaxies Using Self-Organizing Maps*. *Monthly Notices of the Royal Astronomical Society* (2026).
- Hatamnia, H., Mobasher, B., Taamoli, S., **Haghjoo, A.**, et al. *Large-scale Structure in COSMOS-Web: Tracing Galaxy Evolution in the Cosmic Web up to $z \sim 7$ with the Largest JWST Survey*. *Astrophysical Journal* (2026).
- Shuntov, M., Ilbert, O., Lagos, C. del P., **Haghjoo, A.**, et al. *The Stellar Mass Function of Quiescent and Star-Forming Galaxies and its Dependence on Morphology in COSMOS-Web*. *Astronomy & Astrophysics* (2026).
- Yu, S.-Y., Ho, L. C., Tsukui, T., **Haghjoo, A.**, et al. *Through Thick and Thin: The Cosmic Evolution of Disk Scale Height*. *Astrophysical Journal Supplement Series* (2026).
- Hemmati, S., Krick, J., Stern, D., **Haghjoo, A.**, et al. *Reducing the Dimensions of Active Galactic Nuclei Light-Curve Manifolds*. *Astrophysical Journal* (2026).
- Toni, G., Maturi, M., Castignani, G., **Haghjoo, A.**, et al. *COSMOS-Web Galaxy Groups: Evolution of the Red Sequence and Quiescent Galaxy Fraction*. *Astronomy & Astrophysics* (2026).
- Aparicio, V., Jones, D. O., Hoogendam, W. B., **Haghjoo, A.**, et al. *Discovery and Analysis of a Type II Supernova Candidate at $z = 3.19$ from JWST's COSMOS-Web Survey*. Submitted.
- Gozaliasl, G., Yang, L., Kartaltepe, J. S., **Haghjoo, A.**, et al. *Brightest Group Galaxies in COSMOS-Web: Evolution of the Size–Mass Relation Since $z = 3.7$* . *Astronomy & Astrophysics* (2025).

- Asadi, V., Chartab, N., Zonoozi, A. H., **Haghjoo, A.**, et al. *Machine Learning Classification of COSMOS2020 Galaxies: Quiescent versus Star-Forming*. *Astrophysical Journal* (2025).

Research Experiences and Projects

Super-Resolving Low-Resolution Galaxy Spectra with Deep Learning

Dec 2024 – Present

Under the supervision of [Bahram Mobasher](#) and [Shoubaneh Hemmati](#)

UCR

- Developed a generative model to enhance spectral resolution of PRISM data by a factor of 10 using data from the JADES program
- Trained and validated the model with a physics-informed loss function targeting emission line accuracy and flux ratios (e.g., $H\alpha/H\beta$), achieving improved deblending of close spectral features
- Demonstrated the potential of the method to increase the resolution of deep spectral data from future instruments (e.g., Nancy Grace Roman Space Telescope)
- Currently extending the model to a multi-modal foundation framework that integrates photometric and spectroscopic data, enabling cross-modal inference

H2O Mission Catalog

Dec 2024 – Present

Under the supervision of [Bahram Mobasher](#)

UCR

- Inspected 2D spectral data from the Keck Observatory
- Extracted spectroscopically confirmed sources
- Assisted in constructing the H2O catalog, including redshifts and emission lines

Investigating the Effects of Non-Standard Physics on the Global 21cm Signal

Sept 2021 – Dec 2023

Under the supervision of [Jonathan Sievers](#) and [Oscar Hernandez](#)

McGill University

- Investigated the potential of the global 21cm signal to reveal non-standard physics
- Developed a pipeline using Markov Chain Monte Carlo (MCMC) and Levenberg-Marquardt (LM) algorithms to estimate the astrophysical parameters of the global signal
- Employed Accelerated Reionization Era Simulations (ARES) to generate theoretical models
- Validated the approach using data from Experiment to Detect the Global EoR Signature (EDGES)
- Results will aid in constraining future models and defining theoretical limits for upcoming experiments

Honours and Awards

UCR Dean's Fellowship	2024
McGill Faculty of Science Graduate Award	2023
McGill Graduate Excellence Award	2021
Trottier Space Institute at McGill Graduate Fellowship	2021
National University Entrance Examinations: Top 0.5% Among 180K Candidates	2017

Presentations

Deep Learning for Astronomical Data Enhancement

June 2026

Verbal presentation at the [248th AAS Meeting](#)

Pasadena, CA

Learning to see sharper: A physics-informed AI Framework for Spectral Super-Resolution

June 2026

Poster presentation at the [2026 Conference on Physics and AI \(PAI26\)](#)

Stanford University

Learning to see sharper: A physics-informed AI Framework for Spectral Super-Resolution

Apr 2026

Verbal presentation at the [AstroAI Lunch Talks](#) (Invited)

CfA, Harvard and Smithsonian

Learning to see sharper: A physics-informed AI Framework for Spectral Super-Resolution

Mar 2026

Verbal presentation at the [Euclid US Tag up Meeting](#) (Invited)

Euclid Consortium

Super-Resolving Low-Resolution Galaxy Spectra

Jan 2026

Poster presentation at the [Arizona Winter School on Cosmology in the Roman ST Era](#)

University of Arizona

Super-Resolving Low-Resolution Galaxy Spectra

Oct 2025

Verbal presentation at the [IR Spectroscopy from Space](#)

IPAC, Caltech

Super-Resolving Low-Resolution Galaxy Spectra Verbal presentation at the Greater IPAC Science Symposium	Aug 2025 IPAC, Caltech
Super-Resolving Low-Resolution Galaxy Spectra Poster presentation at the Cosmic Cartography with Roman	Jul 2025 Space Telescope Science Institute
Super-Resolving Low-Resolution Galaxy Spectra Verbal presentation at the AstroAI Workshop	Jul 2025 CfA, Harvard and Smithsonian
Unraveling Non-Standard Physics Through the Global 21cm Signal Verbal presentation at the SUT Cosmology Seminar	Jun 2023 SUT
Parameter Estimation of the Global 21cm Signal Poster presentation at Cosmology on Safari 2023	Mar 2023 Hluhluwe, South Africa
Clarifying the Effects of Non-Standard Physics on the Global 21cm Signal Verbal presentation at the McGill Radio Lab Meetings	Oct 2022 McGill University
Effects of Non-Standard Physics on 21cm Signal Verbal presentation at the McGill Early Universe Cosmology Meetings	May 2022 McGill University

Work Experience

Teaching Assistant Department of Physics and Astronomy	2025 – Present UCR
Teaching Assistant Department of Physics	2022 – 2023 McGill University
Teaching Assistant Department of Physics	2019 – 2020 SUT

Technical Skills

Programming Languages	Python, Matlab, $\LaTeX$ Bash, C, C++, Quick Basic, Visual Basic
Platforms	GitHub, GitLab, SLURM

Languages

Persian	Native
English	Fluent / Full Professional Proficiency

Selected Courses

Undergraduate Program	Graduate Program
- Machine Learning in Physics - General Relativity - Quantum Field Theory - String Theory - Elementary Particles - Modeling Statistical Phenomena - Complex Systems	- Computational Physics - Galaxy Formation and Evolution - Astrophysics of the Interstellar Medium - Advanced Cosmology - Observational Astrophysics - Radiative Processes in Astrophysics - Stellar Structure and Evolution

Other Activities and Interests

Mentor at STEM High school, Assisting senior students on their projects
Member, [TEDxAveLorne](#) Curation Team
Representative, [McGill Graduate Association of Physics Students \(MGAPS\)](#)
SUT Mentoring Program Volunteer: Senior-Freshman Buddy Program
Amateur Astronomer: Outreach Observing Nights

Sept 2025 – Mar 2026
Jul 2023 – Dec 2023
Jan 2023 – Dec 2023
2020 – 2021
2015 – 2020

References

Bahram Mobasher
Shoubaneh Hemmati
Jonathan Sievers
Oscar Hernandez
Bahram Mashhoon
Shant Baghrum

Distinguished Professor, UCR
Research Scientist, Caltech
Associate Professor, McGill University
Adjunct Professor, McGill University
Emeritus Professor, University of Missouri
Associate Professor, SUT

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